

Part II

DIFFUSION

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FORWARD DIFFUSION PROCESS

Data-to-noise process

Diffusion models define a *fixed noising process* that gradually destroys data structure (Ho et al., 2020; Sohl-Dickstein et al., 2015; Song et al., 2021). Contrary to flow matching, the forward direction is usually $X_0 \sim p_{\text{data}}$ and $X_T \approx \mathcal{N}(0, I)$. The generative model then learns how to reverse this process.

Gaussian corruption

- ▶ A common continuous-time choice is the variance-preserving SDE: $dX_t = -\frac{1}{2}\beta(t)X_t dt + \sqrt{\beta(t)} dW_t$.
- ▶ Its marginal corruption is explicit:

$$X_t = \alpha(t)X_0 + \sigma(t)\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I) \text{ with } \alpha(t) = \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right) \text{ and } \sigma^2(t) = 1 - \alpha^2(t)$$

- ▶ Equivalently, $q_t(x_t | x_0) = \mathcal{N}(x_t; \alpha(t)x_0, \sigma^2(t)I)$, so any noise level t can be sampled directly.
- ▶ A common non-variance-preserving alternative is the variance-exploding case $X_t = X_0 + \sigma(t)\varepsilon$, where $\alpha(t) = 1$ and the variance grows with t .

REVERSE-TIME GENERATIVE DYNAMICS

Why the reverse process is hard

The forward corruption $q_t(x_t | x_0)$ is known by construction, but it induces the marginal $p_t(x_t) = \int q_t(x_t | x_0) p_{\text{data}}(x_0) dx_0$. Generation should reverse $X_T \sim \mathcal{N}(0, I) \longrightarrow X_0 \sim p_{\text{data}}$, but the true reverse transition $p(x_{t-\Delta t} | x_t)$ depends on the unknown marginal p_t .

Reverse-time SDE

- For a forward SDE $dX_t = f(t, X_t) dt + g(t) dW_t$, the reverse-time dynamics are

$$dX_t = [f(t, X_t) - g^2(t)s_t(X_t)]dt + g(t)d\bar{W}_t, \quad t: T \rightarrow 0, \quad \text{where} \quad s_t(x) = \nabla_x \log p_t(x) \quad \text{is the score.}$$

- This is close in spirit to flow matching: it also defines a learned time-dependent vector field, but with an additional stochastic noise term $g(t)d\bar{W}_t$.
- Intuition: the score points toward regions where the noisy density p_t is larger, hence toward more plausible denoised states.
- The noise term $g(t)d\bar{W}_t$ makes the reverse sampler stochastic. This changes individual trajectories, but not the one-time marginals p_t when the score is exact. Indeed, the probability-flow ODE $dX_t = (f(t, X_t) - \frac{1}{2}g^2(t)s_t(X_t))dt$ has the same marginal densities as the reverse SDE, although its paths are deterministic.

DENOISING, SCORES, AND TWEEDIE'S FORMULA

Marginal score: ideal but inaccessible

The reverse dynamics require

$$s_t(x) = \nabla_x \log p_t(x), \quad p_t(x) = \int q_t(x | x_0) p_{\text{data}}(x_0) dx_0.$$

Since p_t is unknown, the marginal score cannot be computed directly.

Denoising interpretation

Since $X_t = \alpha(t)X_0 + \sigma(t)\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, the conditional score is explicit: $\nabla_{x_t} \log q_t(x_t | x_0) = -\frac{x_t - \alpha(t)x_0}{\sigma^2(t)}$. Averaging over the posterior $p(x_0 | x_t = x)$ gives $s_t(x) = \mathbb{E}[\nabla_{x_t} \log q_t(X_t | X_0) | X_t = x]$.

Tweedie's formula

$$\text{Therefore, } s_t(x) = \frac{\alpha(t) \mathbb{E}[X_0 | X_t = x] - x}{\sigma^2(t)}. \quad \text{Equivalently, } \mathbb{E}[X_0 | X_t = x] = \frac{x + \sigma^2(t)s_t(x)}{\alpha(t)}.$$

Thus learning the score is equivalent to learning the optimal denoiser.

TRAINING OBJECTIVES

Denoising score matching

Train a neural score network by regression on the known conditional score:

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{t, X_0, \varepsilon} \left[\lambda(t) \left\| s_{\theta}(t, X_t) + \frac{\varepsilon}{\sigma(t)} \right\|^2 \right], \quad X_t = \alpha(t)X_0 + \sigma(t)\varepsilon.$$

Noise prediction

In practice, DDPMs often predict the injected noise instead of the score:

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t, X_0, \varepsilon} \left[\|\varepsilon - \varepsilon_{\theta}(t, X_t)\|^2 \right], \quad s_{\theta}(t, x) = -\frac{1}{\sigma(t)} \varepsilon_{\theta}(t, x).$$

Noise-level intuition

Large t corresponds to high noise and coarse semantic structure; small t corresponds to low noise and fine details. Training over all t learns a family of denoisers composed during sampling.

DIFFUSION VS FLOW MATCHING

Aspect	Flow Matching	Diffusion
Direction	Noise $\rightarrow$ data: $X_0 \sim p_0$, $X_1 \sim p_{\text{data}}$.	Data $\rightarrow$ noise during training: $X_t = \alpha(t)X_0 + \sigma(t)\varepsilon$. Generation reverses this path.
Dynamics	Deterministic ODE: $\frac{dX_t}{dt} = v_\theta(t, X_t)$.	Reverse-time SDE: $dX_t = (f(t, X_t) - g^2(t)s_\theta(t, X_t))dt + g(t)dW_t$.
Learned object	Velocity field $v_\theta(t, x) \approx v_t(x)$.	Score $s_\theta(t, x) \approx \nabla_x \log p_t(x)$, or equivalently a denoiser/noise predictor.
Conditional target	Known path velocity $U_t = \partial_t \psi_t(X_0, X_1)$.	Known conditional score $\nabla_{x_t} \log q_t(x_t x_0) = -(x_t - \alpha(t)x_0)/\sigma^2(t)$.
Key identity	$v_t(x) = \mathbb{E}[U_t X_t = x]$.	$s_t(x) = \mathbb{E}[\nabla_{x_t} \log q_t(X_t X_0) X_t = x]$.
Training loss	$\min_\theta \mathbb{E} \ v_\theta(t, X_t) - U_t\ ^2$.	$\min_\theta \mathbb{E} \ s_\theta(t, X_t) + \varepsilon/\sigma(t)\ ^2$.
Sampling	Integrate the learned ODE from $t = 0$ to $t = 1$. Usually deterministic.	Run reverse SDE or DDPM updates from noise to data. Usually stochastic.

TRAINING AND SAMPLING ALGORITHMS

Algorithmic view

Training learns a time-indexed denoiser or score. Sampling starts from Gaussian noise and repeatedly applies the learned reverse dynamics to recover a data-like sample.

Algorithm 1 Training a DDPM noise predictor

Require: data distribution p_{data} , noise schedule $\{\alpha_k\}_{k=1}^T$, model ε_θ

- 1: repeat
 - 2: sample $X_0 \sim p_{\text{data}}$
 - 3: sample $t \sim \text{Unif}\{1, \dots, T\}$
 - 4: sample $\varepsilon \sim \mathcal{N}(0, I)$
 - 5: set $X_t = \sqrt{\alpha_t}X_0 + \sqrt{1 - \alpha_t}\varepsilon$
 - 6: take a gradient step on $\nabla_\theta \|\varepsilon - \varepsilon_\theta(t, X_t)\|^2$
 - 7: until convergence
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Algorithm 1 Ancestral DDPM sampling

Require: trained noise predictor ε_θ

- 1: sample $X_T \sim \mathcal{N}(0, I)$
 - 2: for $t = T, T - 1, \dots, 1$
 - 3: compute $\mu_\theta(X_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(X_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \varepsilon_\theta(t, X_t) \right)$
 - 4: sample $z \sim \mathcal{N}(0, I)$, except $z = 0$ for $k = 1$
 - 5: set $X_{t-1} = \mu_\theta(X_t, t) + \tilde{\sigma}_t z$
 - 6: **return** X_0
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REFERENCES FOR THIS PART I

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-  Sohl-Dickstein, J., Weiss, E. A., Maheswaranathan, N., & Ganguli, S. (2015). **Deep unsupervised learning using nonequilibrium thermodynamics.** *International Conference on Machine Learning*.
-  Song, Y., Sohl-Dickstein, J., Kingma, D. P., Kumar, A., Ermon, S., & Poole, B. (2021). **Score-based generative modeling through stochastic differential equations.** *International Conference on Learning Representations*.